Start
Time based solar irradiance
Signal Generator
Sine wave: 0-1000 W/m²
Saturation Block 0 to 1000
Calculate Solar Heat Gain: 2m²x60%xirradiance
Calculate Temp Rise: Heat/(flow x Cp)
Add Cold Water Temp 20 C + Temp rise
Saturation Block 20 C to 70 C
T_Solar
User Setpoint 45 C
Cold water Temp 20 C
MATLAB Function controller
T_solar > Setpoint ?
Yes
No
Heater OFF mix_ratio = (setpoint-T_cold)/(T_solar-T_cold)
Heater ON mix_ratio = 1
Clamp mix_ratio 0-1
Mixed water Temp = T_solar
Calculate Mixed Temp mix_ratio x T_solar + (1-mix_ratio) x T_cold
calculate Temp Defference setpoint - Mixed Temp
calculate Heater Power dT x flow x Cp
Mixed water Temp
Apply Heater Status Power x heater_on
Switch Block
Integrator
Final Output Temp
Total Energy Convert to kWh
End